Second Higest Frequency Element

Q.1)

```
Class Demo {
    Psvm(){

        int[] arr = { 1, 2, 2, 3, 3, 3, 4, 4 };
        int max = 0;
        int secMax = 0;

        for( int i=0; i<arr.length; i++){
            int count = 0;

            for( int j=0; j<arr.length; j++){
                if(arr[i] == arr[j]){
                    count ++;
                }
            }
            if(count > max){
                secMax = max;
                max = count;
            }
            else if( count > secMax && count < max){
                secMax = count;
            }
        }

        for( int i=0; j<arr.length; i++){
            int count = 0;
            for( int j=0; j<arr.length; j++){
                if(arr[i] == arr[j]){
                    count ++;
                }
            }
            if( count == secMax){
                sop(arr[i]);
            }
    } } }
```

0	1	2	3	2
0	1	2	3	4

Q1] → frequency

0 > 0
1 > 0
sec = 0
max = 1

1 > 0 1 < 1

2 > 1
sec = 1
max = 2

```
class Demo{
        Psvm(){

            int max=0;

            for(int i=0; i< arr.length; i++){
                if(arr[i] > max){
                    max = arr[i];
                }
            }
        int[] freq = new int[max+1];

        for(int i=0; i< arr.length; i++){
            freq[arr[i]]++;
        }
        int fmax=0;  int secmax=0;
        for( int i=0; i< freq.length; i++){
            if( freq[i] > max){
                secmax = max;
                max = freq[i];
            }
            else if( freq[i] > secmax && freq[i] < max){
                secmax = freq[i];
            }
        }

        for(int i=0; i< freq.length; i++){
            if( freq[i] == secmax){
                Sop(i);
            }
        }
}
```

Q.1] HashMap →

```
import java.util.HashMap;
class Demo{
        PSVM(){
            int[]arr = {1,2,2,3,3,3,4,4};

            HashMap<Integer,Integer> hm = new HashMap<>();

            for(int i=0; i<arr.length; i++){
                if(hm.containsKey(arr[i])){
                    hm.put(arr[i], hm.getOrDefault(arr[i],0)+1);
                }
                else{
                    hm.put(arr[i],1);
                }
            }
            int max=0; int secMax=0;
            for(int key: hm.keySet()){
                if(hm.get(key) >max){
                    secMax = max;
                    max = hm.get(key);
                }
            }
            Sop(secMax);
        }
}
```

Q.2] Frequency Greater Than K

Print all elements whose frequency is greater than K

for loop

0 1 2 3 4 5

arr = [1, 1, 2, 2, 2, 3, 3] k = 2

output = 2

```
for class Demo1
        Psvm () {
            int [] arr = {1, 1, 2, 2, 2, 3, 3};
            int k = 2;

            for (int i = 0; i < arr.length ; i++) {
                int count = 0;
                for (int j = 0; j < arr.length ; j++) {
                    if (arr[i] == arr[j]) {
                        count ++;
                    }
                }
                if ( count > k) {
                    sop( arr[i]);
                }
            }
        }
```

1 1 ✓
1 1 ✓
2 2 ✓
2 2 ✓
2 2 ✓

Q2] Frequency :-

0	1	2	3	4	5	6
1	1	2	2	2	3	3

0	2	3	2
0	1	2	3

```
class Demo{
        Public static void main (String[] args){
            int arr = {1,1,2,2,2,3,3};
            int k = 2;
            int max=0;

            for(int i=0; i<arr.length ; i++){
                    if (arr[i] >max){
                            max= arr[i];
                    }
            }

            int[] freq = new int [max+1];

            for(int i=0; i<arr.length ; i++){
                    freq[arr[i]]++;
            }

            for(int i=0; i<freq.length; i++){
                    if (freq[i] >k){
                            sop(i);
                    }
            }
```

freq greater than k.

Q.2] HashMap :-

0	1	2	3	4	5	6
1	1	2	2	2	3	3

1 → 2
2 → 3
3 → 2

```
import java.util.HashMap;
class Demo{
        public static void main(String[]args){

            int arr = {1,1,1,2,2,2,3,3}
            int k = 2;

            HashMap<Integer, Integer> hm = new HashMap<>();

            for(int i=0; i<arr.length; i++){
                if (hm.containsKey(arr[i])){
                    hm.put(arr[i], hm.getOrDefault(arr[i],0)+1);
                }
                else{
                    hm.put(arr[i], 1);
                }
            }

            for (int key : hm.keySet(){
                if( hm.get(key) > k){
                    System.out.println(key);
                }
            }
        }
}
```

1, ~~1+1~~ 1, 1
1, 1+1 = 2
2, 1
2, 1+1 = 2
2, 2+1 = 3
3, 2
3, 3+1 = 2

Q.3] Exactly K frequency Elements
print elements that appear exactly K times

arr = {1, 1, 2, 2, 2, 3} , k=2
output = 1.

0	1	2	3	4	5
1	1	2	2	2	3

1 → 2 — k=2
2 → 3 × k=2
3 → 1 × k=2

```
class Demo{
    public static void main (String [] args){

        int arr[] = {1,1,2,2,2,3};
        int k=2;

        for(int i=0; i<arr.length ; i++){
            int count=0;
            for(int j=0; j<arr.length; j++){
                if arr[i] == arr[j]{
                    count ++;
                }
            }

            if( count == k){
                ~~sop(i);~~ sop(arr[i]);
            }
        }
    }
```

Q.3] → frequency — exactly K frequency element

1	1	2	2	2	3

1 →

0	2	3	1
0	1	2	3
	k=2	k!=2	k!=2

```
class Demo{
    public static void main(String[] args){
        int arr[] = {1,1,2,2,2,3}
        int max=0;
        For(int i=0; i<arr.length; i++){
            if(arr[i] > max){
                max = arr[i];
            }
        }
        int freq[] = new int[max+1];

        For(int i=0; i<arr.length; i++){
            freq[arr[i]]++;
        }
        int K=2;
        for(int i=0; i<freq.length; i++){
            if(freq[i] == K){
                sop(i);
            }
        }
}
```

Q.3] Hashmap → Exactly k frequency elements

| 1 | 1 | 2 | 1 | 2 | 3 |

Output = 1

k = 2

1 → 2
2 → 3
3 → 1

```
import java.util.HashMap;
class Demo{
    Public static void main (String[] args){

    int arr[] = {1, 1, 2, 2, 2, 3};
    int k = 2;

    HashMap<Integer, Integer> hm = new HashMap<>();

    For(int i = 0; i < arr.length ; i++){
        if ( hm.containkey (arr[i] )) {
            hm.put (arr[i], hm.getorDefault (arr[i], 0)+1);
        }
        else{
        hm.put(arr[i] ,1);
        }
    }

    for (int key; hm.keySet (){
            if (hm.get (key) == k) {
                System.out.println (key);
            }
    }
}
```

Q.9] Elements Appearing Only Once.
Print all unique (non-repeating) elements

e.g. arr = [1, 1, 2, 2, 2, 3]
Output = 3.

0	1	2	3	4	5
1	1	2	2	2	3

1 → 2
2 → 3
3 → 1

```
Class Demo{
    public static void main (String [] args){
        int arr = { 1, 1, 2, 2, 2, 3};
        ~~int max = 0;~~
        For (int i=0; i< arr.length; i++){
            int count = 0;
            For (int j=0; j< arr.length; j++){
                if (arr[i] == arr[j]){
                    count ++;
                }
            }
            if (count == 1){
                sop (arr[i]);
            }
        }
    }
}
```

TC = $O(n^2)$
SC =

Q4] Frequency :- Elements Appearing Only once.

| 1 | 1 | 2 | 2 | 2 | 3 |

↓

0	2	3	1
0	1	2	3

```
class Demo{
    public static void main(String [] args){

        int arr[] = {1,1,2,2,2,3};
        int max=0;

        for(int i=0; i<arr.length; i++){
            if(arr[i]> max){
                max= arr[i];
            }
        }

        int freq = new int [max+1];

        for(int i=0; i<arr.length; i++){
            freq[arr[i]]++;
        }

        for(int i=0; i<freq.length; i++){
            if(freq[i]==1){
                sop(i);
            }
        }
```

Q.4] HashMap | Elements Appearing only once :

1	1	2	2	2	3
0	1	2	3	4	5

1 → 2
2 → 3
3 → 1

```
import java.util.HashMap;
Class Demo {
    public static void main (String[] args) {

        int arr[] = {1,1,2,2,2,3};

        HashMap<Integer, Integer> hm = new HashMap<>();

        for (int i=0 ; i< arr.length ; i++) {
            if (hm.containsKey (arr[i])) {
                hm.put (arr[i], hm.getOrDefault (arr[i],0)+1);
            }
            else {
                hm.put (arr[i], 1);
            }
        }

        for (int key ; hm.keySet () {
            if (hm.get (key) == 1) {
                System.out.println (key);
            }
        }
    }
}
```

[1, 2, 2, 3] → [1, 3] Remove duplicates

Print unique no. only.

Q.5]

```
import java.util.*;
class Demo {
        public static void main(String[] args) {

            int[] arr = {1, 1, 2, 2, 2, 3};
            int unique = 0;

            for (int i=0; i<arr.length; i++) {
                int count=0;

                for (int j=0; j<arr.length; j++) {
                    if (arr[i] == arr[j]) {
                        count++;
                    }
                }

                if (count == 1) {
                    unique ++;
                }
            }
            int[] freq = new int [unique];
            int index=0

            for (int i=0; i<arr.length; i++) {
                int count=0;
                for (int j=0; j<arr.length; j++) {
                    if (arr[i] == arr[j]) {
                        count++;
                    }
                }
                if (count == 1) {
                    freq[index] = arr[i];
                    index++;
                }
            }
            System.out.println(Arrays.toString(freq));
        }
}
```

1	2	2	2	3

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* by frequency :-

```
import java.util.*;
class Demo{
        psvm(){

        int arr[] = { 1,1,2,2,3};
        int max=0;

        for (int i=0; i<arr.length; i++){
                if(arr[i] > max){
                        max = arr[i];
                }
        }

        int freq[] = new int [max+1];
        for(int i=0; i<arr.length ; i++){
                freq[arr[i]]++;
        }
        int count=0;
        for(int i=0; i< freq.length, i++){
                if (freq[i] == 1){
                        count++;
                }
        }
        int unique[] = new int [count];
        int index=0;
        for(int i=0; i< freq.length; i++){
                if (freq[i] == 1){
                        unique[index] = i;
                        index++;
                }
        }
```

0	1	3	1
0	1	2	3

1	2	2	3

→ [1, 3]

1 → 1
2 → 3
3 → 1

Print

Q.5 :- HashMap :-

```
import java.util.*;
class Demo {
        Psvm() {

        int[] arr = {1,2,2,2,3};

        HashMap<Integer, Integer> hm = new HashMap<>();

        for(int i=0; i<arr.length; i++){
            if(hm.containkey(arr[i])){
                hm.put(arr[i], hm.getOrDefault
                        (arr[i], 1) +1);
            }
            else{
                hm.put (arr[i], 1);
            }
        }
        int count=0;
        for(int key : hm.KeySet()){
            if (hm.get (key) == 1){
                count++;
            }
        }
        ~~for(int i=0, i<~~
        int[] unique = new int[count];
        for(int i=0; i
```

Q6] Most frequent character in string

e.g.

apple → p

```
class Demo{
    Psvm(){

        String str = "apple";
        int max=0;
        char ch= '';
        For(int i=0; i< str.length; i++){
            int count =0;
            For(int j=0; j< str.length; j++){
                if( str.charAt(i) == str.charAt(j)){
                    count ++;
                }
            if( count > max){
                max= count;
                ch= str.charAt(i);
            }
        }
        sop(ch);
    }
```

a == a p==ax
└1 p==p→1
a==px p==p→2
a==px
a==1x
a==ax

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Q.6] by frequency

1							1																		
0	1	2	3	4	5	6																			26

a → 1
p → 2
l → 1
e → 1

```
class Demo {
    psvm {
        String str = "apple";

        int [] freq = new int [26];

        for (int i=0; i< str.length ; i++) {
            freq[str.charAt(i) - 'a']++;
        }
        int max=0;
        int ch= ' ';

        for (int i=0, i< freq.length; i++) {
            if (freq[i] > max) {
                max = freq[i];
                ch = char (i+'a');
            }
        }
        Sop (ch);
```

Q.6] HashMap:-

a → 1
p → 2
l → 1
e → 1

```
import java.util.*;
class Demo {
        psvm() {

        string str = "apple";


        HashMap<Character, Integer> hm = new HashMap<>();

        for (int i=0; i<str.length; i++) {
            ~~if (hm.con~~
            hm.put(str.charAt(i), hm.getOrDefault
                                (str.charAt(i), 0) + 1);

        }

        int max = 0
        Char ch = ' ';

        for (char key : hm.keySet()) {
                if ( hm.get(key) > max) {
                        max = hm.get(key);
                        ch = key;
                }
        }
        sop(ch);
```

Q.7] Second least frequent element in an Array

Return second least frequent element in an array

arr=[1,1,2,2,2,3]
Output = 1

```
class Demo{
    Psvm(){
        int[]arr = {1,1,2,2,2,3};
        int min = ~~arr[0]~~; arr.length;
        int secondmin = ~~arr[0]~~ arr.length;

        for(int i=0; i<arr.length; i++){
            int count=0;

            for(int j=0; j<arr.length; j++){
                if (arr[i] == arr[j]){
                    count++;
                }
            }
            if (count < min){
                secondmin = min;
                min = count;
            }
            else if (count < secondmin && count > min){
                secondmin = count;
            }
        }
```

i=2
2<6
sec=6
min=2
3<2
1<2
sec=2
min=1

```
for (int i=0, i< arr.length; i++ ){
    int count=0;

    for (int j=0, j< arr.length; j++ ){
            if (arr[i] = = arr[j]) {
                    count ++;
            }
    }
    if ( count = = secondmin){
            sop(arr[i]);
    }
}
```

0	2	3	1
0	1	2	3

Q7] by frequency

```
class Demo{
    Prvm (){
        int[] arr = {1,1,2,2,2,3};
        int max=0;

        for(int i=0; i<arr.length; i++){
            if(arr[i] > max){
                max = arr[i];
            }
        }

        int freq[] = new int [max+1];

        for(int i=0; i<arr.length; i++){
            freq[arr[i]] ++;
        }
        int min = freq.length;
        int secondmin = freq.length;

        for(int i=0; i<freq.length; i++){
            if(freq[i] < min){
                secondmin = min
                min = freq[i];
            }
            else if (freq[i] > secondmin && freq[i] < min){
                secondmin = freq[i];
            }
        }
```

2 < 4
sec = 4
min = 2

3 < 1

1 < 2
sec = 2
min = 1

```
for (int i=0; i< freq.length ; i++) {
        if ( ~~secondmin==~~ freq[i] == secondmin) {
                sop(i);
        }
}
```

Q7 by hashmap :-

```
import java.util.*;
class Demo{
        psvm(){

            int[] arr = {1,1,2,2,2,3};

            HashMap<Integer,Integer> hm = new HashMap<>();

            for(int i=0; i<arr.length; i++){
                if(hm.containsKey(arr[i])){
                    hm.put(arr[i], hm.getOrDefault
                                (arr[i], 1)+1);
                }
                else{
                    hm.put(arr[i], 1);
                }
            }
            int min = arr.length;
            int secondmin = arr.length;

            for(int key : hm.keySet()){
                if(hm.get(key) < min){
                    secondmin = min;
                    min = hm.get(key);
                }
            }
            Sop(secondmin);
        }
```

Q.9) Find Missing Elements Between Two Arrays.
elements present in arr1 but missing in arr2.

arr1 = [1, 1, 2, 2, 2, 3, 7,] arr2 = [1, 1, 2, 2, 3]

Output :- 7

1 → 2
1 → 2
2 → 2
2 → 2
2 → 2
3 → 7
7 → 0

if count = 0; then print

```
int arr1[] = {1, 1, 2, 2, 2, 3, 7}; int arr2[] = {1, 1, 2, 2, 3};
  for(int i=0; i<arr1.length; i++){
      int count = 0;
        for(int j=0; j<arr2.length; j++){
            if(arr[i] == arr[j]){
                count ++;
            }
        }
        if (count == 0){
            sop (arr[i]);
        }
  }
```

Q.10] Find Common Elements (With frequency)

Intersection of two arrays

[1, 2, 2, 3] & [2, 2, 4]

Output: [2, 2]

1	2	2	3

2	2	4

3 2 ✗ 1 2 ✗ 2 2 ✓ count 2 2 ✓ count
3 2 ✗ 1 2 ✗ 2 2 ✓ count 2 2 ✓ count
3 4 ✗ 1 4 ✗ 2 4 ✗ 2 4 ✗

```
int arr1 = {1, 2, 2, 3}

int arr2 = {2, 2, 4}

for(int i=0; i<arr1.length; i++){
    int count=0;
        for(int j=0; j<arr2.length; j++){
            if (arr1[i] == arr2[j]) {
                count++;
            }
        }
    System.out.println(" arr[i] + " _ " + count);
}
```

* assignment No:-8 *

Q.13] Replace Elements with frequency

Convert:

[1, 2, 2, 3]

→ [1, 2, 2, 1]

0	1	2	3
1	2	2	3

→

0	1	2	3
1	2	2	1

1 → 1
2 → 2
2 → 2
3 → 1

```
import java.util.*;
class Demo{
        public static void main(String[]args){

            int[] arr = {1, 2, 2, 3}
            int[] freq = new int[arr.length];

i=0
j=0
            for(int i=0; i<arr.length; i++){
                int count=0;
                    for(int j=0; j<arr.length; j++){
                        if(arr[i] == arr[j]){
                            count++;
                        }
                    }

                    for(int k=0; k<arr.length; k++){
                        freq[k] = count;
                    }
            }
            System.out.println(Arrays.toString(freq));
        }
}
```

Q13] Frequency :-

0	1	2	3
1	2	2	3

$\longrightarrow$

0	1	2	3
1	2	2	1

1 → 1
2 → 2
2 → 2
3 → 1

```
import java.util.*;
class Demo{
    public static void main(String[]args){
      int max=0; int[]arr={1,2,2,3};
      for(int i=0; i<arr.length; i++){
            if (arr[i] > max){
                max=arr[i];
            }
    } int freq[] = new int[max+1];
    for (int i=0; i<arr.length; i++){
            freq[arr[i]]++;
    }

        System.out.println(Arrays.toString(freq));
    }
}
```

Q'13] HashMap →

1	2	2	3

→

1	2	2	1

1 → 1
2 → 2
3 → 1

```
import java.util.*;
class Demo{
        public static void main (String[]args){

            int[] arr= {1,2,2,3};

            HashMap<Integer, Integer> hm = new HashMap<>();

            for(int i=0; i<arr.length; i++){
                if(hm.ContainsKey(arr[i])){
                    hm.put(arr[i], hm.getOrDefault
                                (arr[i], 0) + 1);
                }
                else {
                    hm.put(arr[i], 1);
                }
            }
            System.out.println(hm);
        }
}
```